MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746250
Accessary

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-48411 comprise the complete requirements for Accessary.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746250	Circuit Card Assembly, Accessory

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Identification.- This assembly is a single ended printed circuit board consisting of operational amplifiers, gates, monostable multivibrators, and BCD to binary converter necessary to provide an audio intercom, smoke obscuration circuit, lock on delay, and memory address circuit.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746250.

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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +5 VDC
b. +15 VDC
c. -15 VDC

3.2.1.2 Voltage Divider.- The voltage output or Pin 22 shall be 2.5 VDC.

3.2.1.3 Gate output.- The output at Pin 30 shall be capable of being change to either true or false.

3.2.1.4 Pulse output.- The output at Pin 17 shall be a 40 nanosecond pulse going from Logical "0" to Logical "1".

3.2.1.5 One Second Delay.- The output at Pin 36 shall be capable of being delay 1 second.

3.2.1.6 BCD to Binary Conversion.- This circuit board shall be capable of converting a BCD input to a binary output.

3.2.1.7 Smoke output.- The output at Pin 20 shall be such as to generate obscuration for a limit period of time.

3.2.1.8 Audio Amplifier.- The output on Pin 39 shall have a voltage gain of one (1) and a frequency response of 160 Hz to 10kHz or greater. The output on Pin 21 shall have a gain of one thousand (1000) and a frequency response of 160 Hz to 1600 Hz.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 1174 6250 and the engineering parts list.

3.3 Design and Construction.

3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-48411, Drawing 1174625 Q and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-C-48411.

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3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-C-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power supply (PS 1).- The power supply shall be 10 ± 0.5 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power supply (PS_2) & (PS_3).- The power supply shall be 20 ± 1.0 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.3 Oscilloscope.- Rise time of scope plus associated measuring probe of 50 nanseconds or less, sweep period of 50 nanseconds per centimeter of greater, input resistance greater than 8 megohms, input capacitance less than 25 picafarads.

4.2.1.4 Dummy loads.- The dummy loads shall be one 3-input positive "Nand" gate found in an SN54L10, 5410 or equivalent, 10K ohm, ¼W 1% and 50 ohm, 1W 5% resistors.

4.2.1.5 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

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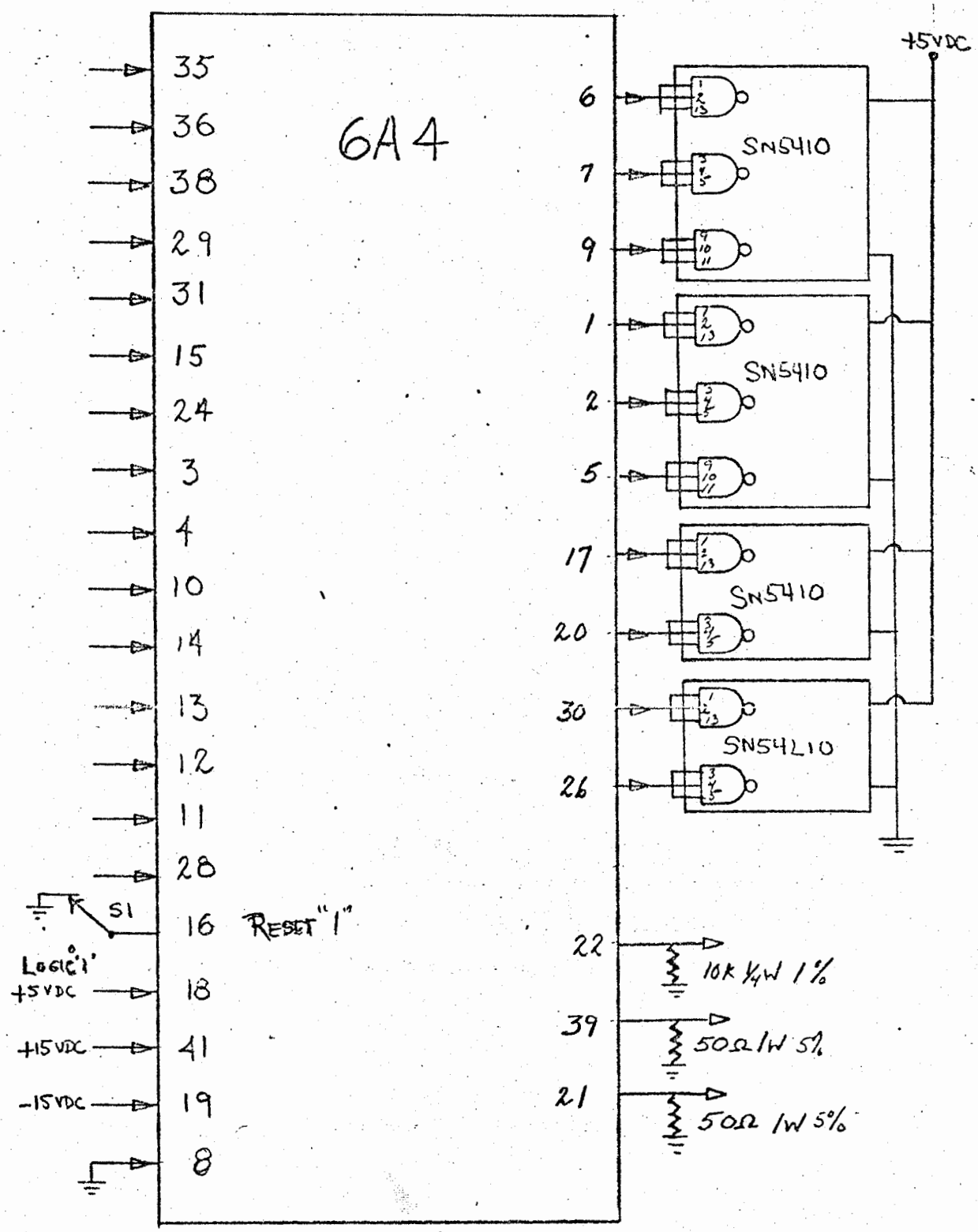


FIGURE 1

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4.2.2.1 Power supply.- All the requirements shall be met with the following power supply voltages.

a. +5 ± 0.05 VDC
b. +15 ± 0.05 VDC
c. -15 ± 0.05 VDC

4.2.2.2 Digital Output.- The "0" or false output level shall be between 0.00 and 0.40 VDC. The "1" or true output level shall be between +2.40 and +5.00 VDC.

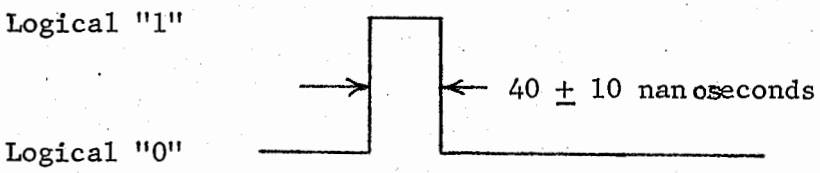
4.2.2.3 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.4 Voltage Divider.- When +5.00 VDC is applied to Pin 18, the output at Pin 22 shall be 2.50 VDC.

4.2.2.5 Gate Output.- When Logic is applied to Pin 28 the corresponding results on Pin 30 shall be as shown below.

Pin 28	Pin 30
Logical "1"	Logical "0"
Logical "0"	Logical "1"

4.2.2.6 Pulse output.- When Reset "1" is applied to Pin 16 the output on Pin 17 shall be as shown below.



4.2.2.7 One Second Delay.- With the inputs as shown in Column A the corresponding output shall be as shown below. When the output is a Logical "1" it shall remain for 1.0 ± .2 seconds. Before reverting to Logical "0".

Column A	Column B
Pin 24	Pin 26
Logical "1"	Logical "0"
Logical "0"	Logical "1"
Logical "1"	Logical "1"

4.2.2.8 BDC to Binary Conversion.- When the Logical input are applied the corresponding output shall be as shown in the Table I below.

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TABLE I

Pins Logic	INPUT					
14	13	12	11	10	4	3
0	0	0	0	0	0	0
0	1	0	0	0	0	1
1	0	0	0	0	1	0
1	0	1	0	1	0	1
0	1	0	1	0	1	0

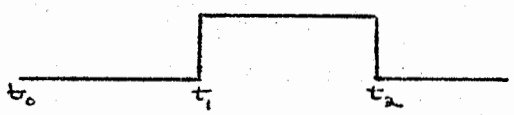
Pins Logic	OUTPUT					
6	7	9	1	2	5	
0	0	0	0	0	0	
0	1	0	1	0	1	
1	0	1	0	1	0	
1	0	0	1	1	1	
0	1	0	1	1	0	

4.2.2.9 Smoke output.- When Logic is applied to Pin 15 the output shall be as shown below of Pin 20.

Input	Output
Pin 15	Pin 20
Logical "0"	Logical "0"
Logical "1"	From Logical "0" to Logical "1" To Logical "0" in 1.40 ± .1 seconds.

t_1-t_0= 0.45 ± 0.05 seconds

t_2-t_1= 0.95 ± 0.05 seconds



4.2.2.10 Audio Output.- When Pins 29 and 31 are grounded and a 750 ± 1 Hz, 1 ± 0.1V p-p sine wave is applied to Pin 38 the output at Pin 39 shall be a 750 ± 1Hz, 1 ± 0.3V p-p sine wave. When the input frequency is varied the output will be 0.5V p-p at a frequency of 160 ± 100 Hz and will be 1V p-p at 10kHz or greater. When Logical "0" is applied to Pins 29 and 31 and a 750 ± 1Hz, 20 ± 1MV p-p sine wave is applied to Pin 36. The output at Pin 21 will be 750 ± 1Hz, 20 ± 3V p-p sine wave. Vary the input frequency at Pin 36 and the output will drop to 10 ± 3V p-p at the frequency of 160 ± 100 Hz and 160 ± 300 Hz.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11745214.

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6. NOTES

6.1 The notes specified in the general specification MIL-C-48411 are applied to this specification.

Custodian:

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Preparing Activity:

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Project No. 1240-A599

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